Cell Tuition
1-1 PRIVATE TUTORS

# Probability as Fractions, Decimals and Percentages

**Name** _____ **Date** _____ No calculator Version 1

**KEY WORDS** probability · outcome · event · equally likely · fraction · complement

## A Probability as a fraction ●○○ ANSWER ON THE SHEET

**Remember:** Probability $= \dfrac{\text{number of outcomes you want}}{\text{total number of outcomes}}$, when every outcome is equally likely.

**1.** A fair dice is rolled. Write the probability of each event as a fraction in its simplest form.

**a)** Rolling a 4 _____ **b)** Rolling an even number _____

**c)** Rolling a number greater than 4 _____ **d)** Rolling a 7 _____

**e)** Rolling a number less than 7 _____ **f)** Rolling a 1 or a 2 _____

**2.** A bag holds 3 red, 5 blue and 2 green counters. One counter is taken out without looking. Write the probability of each event as a fraction in its simplest form.

**a)** A red counter _____ **b)** A blue counter _____ **c)** A green counter _____

## B Probability as a decimal and a percentage ●●○ WORK IN YOUR BOOK

**Remember:** A probability can be written as a fraction, a decimal or a percentage: $\frac{1}{4} = 0.25 = 25\%$.

**3.** A fair spinner is numbered 1 to 10. Write each probability as a fraction, a decimal and a percentage.

**a)** The number 3 **b)** An even number **c)** A number greater than 7

**4.** Cards numbered 1 to 20 are shuffled and one card is picked. Write each probability as a fraction, a decimal and a percentage.

**a)** The number 7 **b)** An odd number **c)** A number greater than 15

## C Events with more than one outcome ●●○ WORK IN YOUR BOOK

**Watch out:** Count every outcome that fits the event. For "a multiple of 3" on a spinner numbered 1 to 10, the outcomes 3, 6 and 9 all count.

**5.** A fair spinner is numbered 1 to 10. Find the probability of each event, as a fraction in its simplest form.

**a)** A multiple of 3 **b)** A prime number **c)** A number less than 4

**d)** A factor of 10 **e)** An odd number **f)** A number greater than 10

**6.** A bag holds 5 red, 4 blue, 2 green and 1 yellow counters. One counter is taken out without looking. Find the probability that it is:

**a)** red or blue **b)** green or yellow **c)** not red


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## D Complementary events ●●● WORK IN YOUR BOOK

> **Remember:** An event either happens or it does not, so the two probabilities add to 1 (or 100%). If the probability of rain is 0.3, the probability of no rain is $1 - 0.3 = 0.7$.

**7.** The probability that an event happens is given. Find the probability that it does not happen.

**a)** $0.3$ **b)** 35% **c)** $\frac{3}{8}$ **d)** $0.9$

**e)** 80% **f)** $\frac{1}{6}$ **g)** $0.25$ **h)** $\frac{2}{5}$

## E Comparing likelihood ●●● WORK IN YOUR BOOK

**8.** Which is more likely? Show your working.

**a)** rolling a 6 on a fair dice, or picking a red counter from a bag with 2 red and 8 blue counters?

**b)** rolling an even number on a fair dice, or getting a head on a fair coin?

**c)** picking a vowel from the letters of MATHS, or picking a vowel from the letters of NUMBER?

## ★ Extension FOR EARLY FINISHERS

**9.** A bag holds 10 counters. The probability of picking a red counter is $\frac{2}{5}$. How many red counters are in the bag?

**10.** The probability of winning a game is 0.2. Tama says, "So if I play 5 times I am certain to win once." Is he right? Explain.



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# Probability as Fractions, Decimals and Percentages

**Name** _____ **Date** _____ No calculator **Version 2**

**KEY WORDS** probability · outcome · event · equally likely · fraction · complement

## A Probability as a fraction ●○○ — ANSWER ON THE SHEET

**Remember:** Probability $= \dfrac{\text{number of outcomes you want}}{\text{total number of outcomes}}$, when every outcome is equally likely.

**1.** A fair dice is rolled. Write the probability of each event as a fraction in its simplest form.

**a)** Rolling a 3 _____ **b)** Rolling an even number _____

**c)** Rolling a number greater than 2 _____ **d)** Rolling a 7 _____

**e)** Rolling a number less than 7 _____ **f)** Rolling a 1 or a 2 _____

**2.** A bag holds 4 red, 5 blue and 1 green counters. One counter is taken out without looking. Write the probability of each event as a fraction in its simplest form.

**a)** A red counter _____ **b)** A blue counter _____ **c)** A green counter _____

## B Probability as a decimal and a percentage ●●○ — WORK IN YOUR BOOK

**Remember:** A probability can be written as a fraction, a decimal or a percentage: $\frac{1}{4} = 0.25 = 25\%$.

**3.** A fair spinner is numbered 1 to 10. Write each probability as a fraction, a decimal and a percentage.

**a)** The number 3 **b)** An even number **c)** A number greater than 6

**4.** Cards numbered 1 to 20 are shuffled and one card is picked. Write each probability as a fraction, a decimal and a percentage.

**a)** The number 7 **b)** An odd number **c)** A number greater than 16

## C Events with more than one outcome ●●○ — WORK IN YOUR BOOK

**Watch out:** Count every outcome that fits the event. For "a multiple of 3" on a spinner numbered 1 to 10, the outcomes 3, 6 and 9 all count.

**5.** A fair spinner is numbered 1 to 10. Find the probability of each event, as a fraction in its simplest form.

**a)** A multiple of 4 **b)** A prime number **c)** A number less than 4

**d)** A factor of 10 **e)** An odd number **f)** A number greater than 10

**6.** A bag holds 4 red, 3 blue, 3 green and 2 yellow counters. One counter is taken out without looking. Find the probability that it is:

**a)** red or blue **b)** green or yellow **c)** not red



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## D Complementary events ●●● WORK IN YOUR BOOK

**Remember:** An event either happens or it does not, so the two probabilities add to 1 (or 100%). If the probability of rain is 0.3, the probability of no rain is $1 - 0.3 = 0.7$.

**7.** The probability that an event happens is given. Find the probability that it does not happen.

**a)** $0.4$ **b)** 45% **c)** $\frac{5}{8}$ **d)** $0.7$

**e)** 60% **f)** $\frac{5}{6}$ **g)** $0.75$ **h)** $\frac{3}{5}$

## E Comparing likelihood ●●● WORK IN YOUR BOOK

**8.** Which is more likely? Show your working.

**a)** rolling a 1 on a fair dice, or picking a red counter from a bag with 3 red and 9 blue counters?

**b)** rolling an odd number on a fair dice, or getting a tail on a fair coin?

**c)** picking a vowel from the letters of GRAPH, or picking a vowel from the letters of RANDOM?

## ★ Extension FOR EARLY FINISHERS

**9.** A bag holds 12 counters. The probability of picking a red counter is $\frac{1}{4}$. How many red counters are in the bag?

**10.** The probability of winning a game is 0.2. Tama says, "So if I play 5 times I am certain to win once." Is he right? Explain.


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